nature materials

Review article

https://doi.org/10.1038/s41563-024-01989-8

Towards the scalable synthesis of two-dimensional heterostructures and superlattices beyond exfoliation and restacking

Received: 2 February 2024

Accepted: 2 August 2024

Published online: 3 September 2024

Check for updates

Jia Li [1,5], Xiangdong Yang [2,5], Zhengwei Zhang [3], Weiyou Yang [2], Xidong Duan [1] ✉ & Xiangfeng Duan [4] ✉

Two-dimensional transition metal dichalcogenides, which feature atomically thin geometry and dangling-bond-free surfaces, have attracted intense interest for diverse technology applications, including ultra-miniaturized transistors towards the subnanometre scale. A straightforward exfoliation-and-restacking approach has been widely used for nearly arbitrary assembly of diverse two-dimensional (2D) heterostructures, superlattices and moiré superlattices, providing a versatile materials platform for fundamental investigations of exotic physical phenomena and proof-of-concept device demonstrations. While this approach has contributed importantly to the recent flourishing of 2D materials research, it is clearly unsuitable for practical technologies. Capturing the full potential of 2D transition metal dichalcogenides requires robust and scalable synthesis of these atomically thin materials and their heterostructures with designable spatial modulation of chemical compositions and electronic structures. The extreme aspect ratio, lack of intrinsic substrate and highly delicate nature of the atomically thin crystals present fundamental difficulties in material synthesis. Here we summarize the key challenges, highlight current advances and outline opportunities in the scalable synthesis of transition metal dichalcogenide-based heterostructures, superlattices and moiré superlattices.

Human civilization has challenges prominently shaped by innovations in materials, marking distinct eras characterized by the critical materials utilized during each period—such as the Stone Age, the Bronze Age and the Iron Age. The invention of silicon-based integrated circuits in the 1960s signified the beginning of the Silicon Age[1,2]. Over the past half-century, we have witnessed enormous progress (and profits) in silicon-based integrated circuits, which define the material foundation for the information technology revolution and the rapid expansion of artificial intelligence. This progress in silicon-based integrated circuits has been primarily driven by sustained miniaturization, facilitated by continued technological innovations in fabrication technologies and fundamental breakthroughs in device architecture designs[3]. However, despite (and because of) these extraordinary advances, further scaling of silicon-based electronics is facing increasing challenges due to the rapid degradation of electronic properties (such as carrier mobility) in silicon approaching atomic thickness (for example, sub-5 nm or sub-3 nm regimes), which has motivated intensive efforts to find alternative materials that can meet the future demands of computing power[4,5].

A full list of affiliations appears at the end of the paper. ✉e-mail: xidongduan@hnu.edu.cn; xduan@chem.ucla.edu

In this regard, two-dimensional atomic crystals (2DACs), particularly layered transition metal dichalcogenides (TMDs), have received considerable interest[6,7]. These crystals feature atomically thin thickness (monolayer thickness <1 nm) and dangling-bond-free surfaces, unique attributes that allow 2DACs to retain excellent electronic properties (for example, carrier mobility) at the extreme physical scale, making them attractive candidates for ultimate transistor scaling beyond the reach of silicon technology[8]. Furthermore, with these atomically thin geometries and widely variable chemical compositions and physical properties (for example, semiconductors, superconductors, metals, Weyl semimetals, topological insulators and ferromagnets), 2D TMDs also offer an exciting materials platform for exploring exotic physical properties at the limit of atomic thickness[9,10]. The rich family of 2D TMDs could also allow atomic-scale integration of highly distinct chemical compositions and electronic structures to create diverse heterostructures. This could enable elaborate modulation of the potential landscape to control charge generation, separation, transport, correlation or recombination and rationally tailor their electronic, optical and emergent quantum properties by design[11,12]. As a result, 2DACs have been the most intensively studied materials system in the context of emerging electronic materials in the past decade.

Many of these atomically thin 2D TMDs and heterostructures are typically prepared through a straightforward mechanical exfoliation-and-restacking process[13]. This method has greatly contributed to the rapid expansion of the field, leading to many exciting discoveries of exotic physical phenomena and proof-of-concept demonstrations of new devices with exceptional performance or unprecedented functions[11,14]. However, despite these extraordinary advancements, the scalability and practicality of exfoliated flakes and restacked heterostructures for real-world technologies remain challenging. While the accessibility of exfoliated 2D flakes and restacked heterostructures has greatly accelerated initial progress in the field, it has ironically slowed efforts towards the rational synthesis of 2D materials and their heterostructures, particularly in the early stages of research.

Unlike previous research in new materials systems (for example, semiconductors and their heterostructures, quantum dots, carbon nanotubes or semiconductor nanowires) where fine synthetic control is a prerequisite for systematically investigating physical properties investigation and meaningful device demonstrations, the early research in 2D materials largely bypassed/evaded the material development phase. While there have been increasing efforts and notable recent progress in material growth, synthetic control of this unique materials system generally lags behind the overall development of the field, posing a limiting factor on further advancement[15]. Unlocking the full potential of 2D TMDs necessitates controllable synthesis of large single crystals, as well as their heterostructures and superlattices, with precise control over the spatial modulation of chemical compositions and electronic structures[16,17]. This is akin to the approach taken in traditional semiconductor heterostructures and superlattices, which define the fundamental materials foundation for all modern electronic and optoelectronic devices. In this Review, we aim to discuss the essential challenges, recent advances and our perspectives on the synthetic control of 2D TMDs, their heterostructures and superlattices.

Fundamental challenges in material synthesis

With thicknesses measured in single (or few) atoms and lateral dimensions extending to a macroscopic scale (for example, 300 mm for a 12-inch wafer), 2D materials exhibit an extreme aspect ratio of up to 10^9 or larger. This exceptional aspect ratio necessitates highly anisotropic growth, during which growth rates differ by millions or billions of times along the lateral (parallel to the 2D basal plane) and vertical (normal to the 2D basal plane) dimensions, posing a fundamental synthetic challenge. The chemical vapour deposition (CVD) process stands out as one of the most compelling growth methods for the scalable production of 2DACs. This method is synergistically controlled by factors such as the vapour pressure and temperatures of the precursor materials, as well as substrate properties and temperature.

Owing to the lack of an intrinsic substrate for most 2D TMDs, growth is typically carried out on a foreign substrate with a chemical composition and lattice structure that are distinct from the material of interest. Consequently, the nucleation and growth processes of 2D TMDs fundamentally differ from conventional vertical epitaxial growth on lattice-matched substrates. Generally, the growth of 2D crystals is initiated through either homogeneous nucleation in the vapour phase or heterogeneous nucleation on a foreign substrate. When the vapour pressure of the chemical precursors surpasses a certain threshold, small 2D nuclei may begin to form in the vapour phase and subsequently settle onto the substrate. Alternatively, nuclei may directly form on the surface of the chosen substrate, where specific surface defects can act as preferred nucleation sites. In this case, precise control of surface defects could facilitate tailoring of the nucleation process. Once a nucleus forms on the substrate surface, growth may proceed either laterally with a given lateral growth rate (R_L), facilitating the continued extension of nuclei in the lateral dimension to produce large-area 2D TMDs, or vertically with a certain vertical growth rate (R_V) resulting in the production of multi-layered materials.

Under a given condition (temperature, vapour pressure, atmosphere field and substrate), several competing processes may occur (Fig. 1a): (1) the homogeneous formation of nuclei in the vapour phase that eventually settle on the bare substrate to form a new growth centre ($R_V > 0$) or on top of existing 2DACs ($R_V > 0$); (2) the heterogeneous formation of new nuclei on the bare substrate or top of the existing 2DACs ($R_V > 0$); (3) the addition of atoms to the growth front (edge) of the existing nuclei ($R_L > 0$); (4) vaporization of the constituent atoms at the edge of existing 2D TMDs (negative lateral growth, $R_L < 0$); and (5) vaporization in the basal plane of the 2D TMDs (negative vertical growth, $R_V < 0$). Growing uniform 2D TMD thin films requires delicate control of oversaturation that is sufficient for the continued addition of the precursor atoms to the edge (growth front) of the existing 2D domains ($R_L > 0$) but insufficient for homogeneous or heterogeneous nucleation of new 2DAC nuclei ($R_V = 0$). In this regard, there is an intrinsic trade-off: supersaturation that is too low could lead to very low lateral growth, whereas supersaturation that is too high could lead to the nucleation of new 2D domains and poor control of the uniformity of the thickness.

In a practical growth apparatus, slight variations in the local conditions (for example, the temperature gradient or vapour pressure gradient) (Fig. 1b,c) could also lead to spatial fluctuations in supersaturation at different locations, resulting in a positive growth rate in one location but a negative growth rate in another. Such negative growth can be a particular concern for atomically thin 2D TMDs, which often consist of volatile chalcogen elements that are prone to desorption/evaporation and are highly susceptible to even the slightest fluctuations in the chemical environment. A minute change in the synthetic conditions, such as the local vapour pressure, substrate temperature or flow dynamics, could lead to undesirable vacancies that could compromise the electronic properties or a negative growth rate that may rapidly consume (etch away) the entire thickness of the monolayer crystals (Fig. 1a).

Ensuring the uniform growth of monolayer TMDs therefore requires precise control of the growth rate in the vertical or lateral directions by regulating key thermodynamic and kinetic factors (Fig. 1c). This is particularly challenging for large-area growth. To guarantee stable growth over a large area, an excessive supersaturation is usually required to provide efficient and continued lateral growth

https://doi.org/10.1038/s41563-024-01989-8

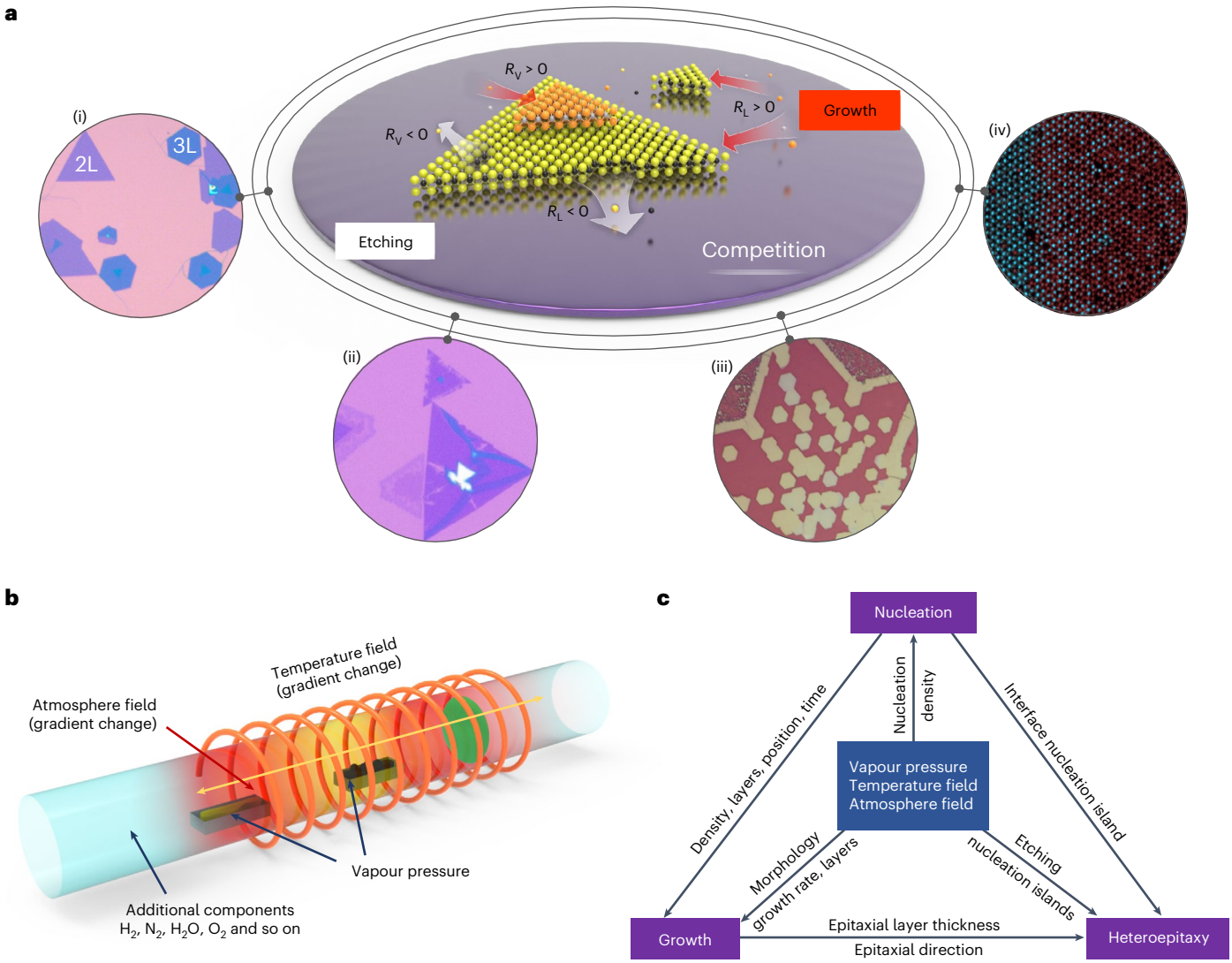

Fig. 1 | Challenges in CVD synthesis of 2D TMDs and heterostructures. a, Four synthetic challenges for 2DACs and their heterostructures are shown: (i) competition between in-plane and out-of-plane growth (2L, bilayer; 3L, three layer); (ii) uncontrolled competition between growth and etching; (iii) random and uncontrolled nucleation; and (iv) non-ideal heterointerfaces. **b**, Schematic diagram of the CVD system for preparing wafer-scale 2DACs. **c**, The processing parameters that affect nucleation, growth and heteroepitaxy of 2DACs and their heterostructures. Panel **a** reproduced with permission: (i), ref. 22, American Chemical Society; (ii), ref. 16, AAAS. Panel **a**(iv) adapted with permission from ref. 46, Springer Nature Limited.

across the entire area, which may lead to the unavoidable nucleation of new crystal layers and vertical growth in some locations, and thus a non-uniform thickness distribution (Fig. 1a). Furthermore, the growth of heterostructures or superlattices requires a switch in the chemical environment or processing conditions (thus the supersaturation) that may result in a negative growth rate (lateral or vertical etching) in the existing TMD layers (Fig. 1a), making the robust growth of heterostructures a persistent challenge. Depending on growth conditions, the resulting heterostructure could also be plagued by atomic diffusion, ion exchange and atom replacement, resulting in heterostructures with poorly controlled interfaces (Fig. 1a).

Despite a slow start, we have seen a tremendous recent surge in efforts to achieve synthetic control of 2D TMDs. Here we consider these efforts in four closely correlated topics: (1) CVD synthesis of large-area monolayer TMD films and single crystals; (2) synthesis of lateral heterostructures and superlattices; (3) synthesis of vertical heterostructures and moiré superlattices; (4) synthesis of vertical superlattices; and (5) site-selective growth. We conclude with an overview of potential future challenges and directions in the synthetic control of 2D TMD materials.

CVD synthesis of large-area monolayer TMD films and single crystals

Over the past 5–10 years, substantial efforts have been devoted to precisely controlling the nucleation and growth kinetics for the scalable CVD synthesis of TMDs with tailored thicknesses, single-crystal sizes and crystalline phases to systematically modulate their electronic, chemical, quantum and other emerging properties. Depending on the precursor materials, the synthetic processes can be classified into three main categories: (1) chalcogenization of transition metal or metal oxide thin films, (2) solid-source thermal CVD and (3) metal–organic CVD (MOCVD) processes. Chalcogenization of transition metal or metal oxide thin films typically involves the physical vapour deposition (PVD) of a transition metal (for example, Mo) or transition metal oxide (such as MoO_3) thin film followed by a high-temperature sulfurization/selenization reaction[18,19]. Benefiting from well-developed and scalable PVD processes, wafer-scale and patterned TMD thin films can be readily produced using this approach[20]. Solid-source thermal CVD processes generally involve the vaporization of separate solid sources of metal precursors, chalcogen precursors or the bulk solid

TMD itself, generating vapour-phase reactants that are transported by a carrier gas to the growth substrate, where they nucleate and grow into atomically thin TMD crystals[21,22]. This process is relatively versatile and represents the most widely studied method for producing 2D TMDs so far[23]. With metal–organic vapour-phase precursors, the MOCVD approach is commonly used in the compound semiconductor industry to produce semiconductor thin films, heterostructures and superlattices[24,25]. To enhance the uniformity and controllability of large-area growth, several strategies, including controlling the supply of metal or non-metal precursors (for example, multisource designs[26], 'face tof ace'[27]) and engineering the substrate[28], have been explored to control nucleation density and stitching processes. Motivated by these improvements, considerable progress has been achieved in the production of wafer-scale monolayer MoS_2 films with sizes ranging from 2 to 12 inches (refs. 29,30).

Monolayer TMD thin films grown by CVD typically feature polycrystalline structures with abundant grain boundaries that can greatly compromise the overall quality and uniformity of their electronic properties. Synthesizing single-crystal 2D films on the wafer scale is of paramount importance to ensure device-to-device uniformity in practical technologies. There are two feasible routes to obtain large-scale single-crystal 2DACs: single-seed lateral growth at the wafer scale and oriented multi-seed growth and seamless stitching. Large-area $MoSe_2$ with a single-crystal domain size of up to 2.5 mm was grown on a molten glass substrate[31] and 1 inch single-crystal 2H $MoTe_2$ has been realized by a single-seed-induced phase transition and recrystallization process[32]. A larger-scale approach involves the oriented growth of multiple domains on a single-crystal substrate that then seamlessly merge into wafer-scale single crystals[33]. The unidirectional growth of multiple domains greatly depends on the surface symmetry of the substrate and the 2D materials, and the coupling interaction between them[34]. Two typical strategies have so far proved to be effective in realizing unidirectional growth of TMD domains: step-edge-guided epitaxy[35] and surface-symmetry-guided epitaxy[36]. The step edges can lower the surface symmetry and break the energy degeneracy of antiparallel domain grains, enabling unidirectional alignment of TMD domains that are conducive to large-area single-crystal formation[37,38]. For example, *c*-plane sapphire wafers with a major miscut angle, creating surface steps of 0.4 nm in height along $\langle 10\bar{1}0\rangle$, allow unidirectional growth of MoS_2 domains, finally producing 2 inch monolayer MoS_2 single crystals with wafer-scale uniformity[33]. Similarly, 2 inch single-crystal WS_2 monolayers were grown on vicinal *a*-plane sapphire surfaces[39]. It has also recently been suggested that the interaction of 2D TMD grains with an exposed oxygen–aluminium atomic plane, rather than the exposed step edges, is a more dominant factor enabling such unidirectional MoS_2 domain growth on sapphire substrates[36].

Synthesis of 2D lateral heterostructures and superlattices

With a highly similar structure and relatively small lattice mismatch, TMD–TMD heterostructures can be produced through a lateral epitaxial process[40,41]. For example, a two-step thermal CVD process with in situ switching of chemical vapour supplies allows the synthesis of WS_2–WSe_2 and MoS_2–$MoSe_2$ lateral heterostructures[42]. Specifically, to produce a WS_2–WSe_2 heterostructure, the WS_2 domains were first grown on a SiO_2 substrate by thermally vaporizing a solid WS_2 source; next, in situ switching of the chemical vapour source to WSe_2 without exposing the WS_2 domain to ambient conditions enabled the continued lateral heteroepitaxial growth of WSe_2 at the peripheral active growth front to form WS_2–WSe_2 lateral heterostructures. In parallel, a series of two-block TMD lateral heterostructures with specific combinations including $MoSe_2$–WSe_2 (ref. 43), WS_2–MoS_2 (ref. 44) and MoS_2–WSe_2 (ref. 45) have been successfully realized by one-step or two-step CVD methods.

Such lateral epitaxial growth processes can, in principle, be repeated multiple times to enable the block-by-block growth of multi-heterostructures or superlattices. However, 2D crystals comprising just a single atomic layer or a few atomic layers are generally highly delicate with little tolerance to changes in synthetic conditions at high temperatures. The repeated switching of distinct chemical vapour sources and/or growth conditions in the sequential growth of multi-heterostructures or superlattices could easily cause serious degradation or complete etching of the atomic layers. The growth of atomically thin 2D heterostructures is therefore fundamentally much more challenging than that of other bulk or nanoscale heterostructures with a greater number of atomic layers. To address this challenge, a modified thermal CVD process (Fig. 2a) was designed with a reverse Ar flow from the substrate to the source to cool and protect the existing block during the temperature swing stage between sequential growth steps[16]. In this process, a forward flow from the chemical vapour source was only applied at the specific growth temperature and conditions. The reverse flow continually flushed the existing monolayer materials on the growth substrate with cold Ar gas during the temperature swing to reduce exposure to high temperatures and minimize thermal degradation. The reverse flow during the temperature ramping-up stage could also prevent an unintended supply of the chemical vapour source at an undesired temperature and eliminate uncontrolled homogeneous nucleation. With this high degree of controllability at each step, the integrity and quality of monolayer heterostructures can be preserved after multiple sequential growth steps, establishing a general and reliable strategy for the growth of a wide range of heterostructures, multi-heterostructures and superlattices (Fig. 2b–d) with atomically sharp and clean interfaces (Fig. 2e).

In parallel, a one-pot growth method was reported for the growth of 2D lateral superlattices[46]. In this case, a mixed solid source composed of MoX_2 and WX_2 is used, and the selective growth of each TMD is controlled by the carrier gas (Fig. 2f). A mix of N_2/H_2O promotes the growth of MoX_2; while Ar/H_2 promotes the growth of WX_2 and inhibits the growth of MoX_2. Thus superlattices can be synthesized by simply switching the carrier gas during growth without exchanging the solid source or reactor (Fig. 2g–i). However, it is difficult to instantly switch the chemical source and thus the resulting heterostructures generally exhibit an alloyed interface, instead of an atomically sharp interface.

The MOCVD process has also been explored for synthesizing coherent lateral WS_2–WSe_2 superlattices (Fig. 2j,k). Compared with solid-source CVD, MOCVD offers intrinsic advantages in terms of better control over the introduction of multiple chemical precursors and rapid switching between them. Remarkably, the width of each component in the superlattice can be accurately tuned to nanoscale dimensions, enabling the precise engineering of 2D heterostructures and superlattices with tunable strain and optical properties[47].

Beyond the typical concentric heterostructures or superlattices, the growth of monolayer TMD lateral heterostructure arrays represents a crucial step towards the adoption of 2D heterostructures in integrated electronic and photoelectronic device arrays. The heterostructure arrays can be realized by combining lithographic patterning with lateral epitaxy growth processes. To this end, the scalable preparation of 2DAC templates with atomic precision is an important prerequisite for producing monolayer lateral heterostructure arrays. However, conventional lithography usually results in a relatively rough edge configuration that is incompatible with heterostructures with atomically sharp interfaces and can easily cause undesired residues or damage on the surfaces of 2DACs, which may function as undesired nucleation sites for highly heterogeneous growth of multilayer materials. To address this challenge, we developed a laser patterning and anisotropic thermal etching process to produce patterned 2DAC arrays for the scalable growth of monolayer mosaic heterostructure (MMH) arrays[48] (Fig. 2l). In this method, controlled thermal etching at point defects created by a laser-patterning process produces periodic triangle holes

https://doi.org/10.1038/s41563-024-01989-8

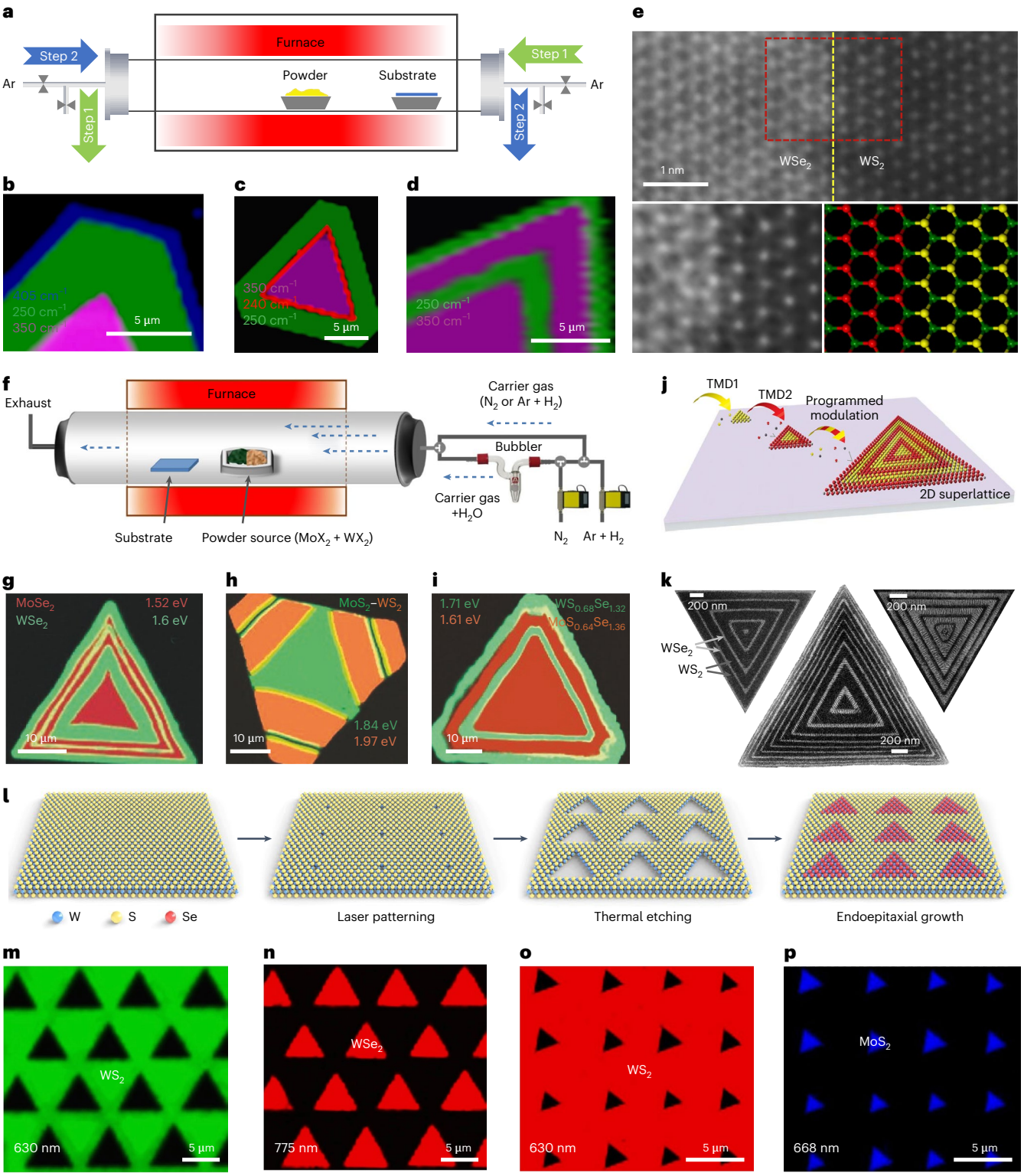

Fig. 2 | Synthesis of 2D lateral heterostructures and superlattices. a, Schematic illustration of a modified CVD system for the robust epitaxial growth of lateral heterostructures and superlattices. **b**–**d**, Raman mapping of synthesized WS_2–WSe_2–MoS_2 (**b**) and WS_2–$MoSe_2$–WSe_2 (**c**) lateral heterostructure and a WS_2–WSe_2–WS_2–WSe_2–WS_2 lateral superlattice (**d**). The values 350 cm^{-1}, 250 cm^{-1}, 405 cm^{-1} and 240 cm^{-1} are characteristic Raman peaks of WS_2, WSe_2, MoS_2 and $MoSe_2$, respectively. **e**, High-resolution STEM images of a WS_2–WSe_2 heterostructure (top). The zoom-in STEM image marked in the red box in top panel (bottom left) and the corresponding atomic model diagram (bottom right)[16]. **f**, Schematic illustration of a one-pot CVD system for the growth of lateral superlattices. **g**–**i**, Photoluminescence mapping of $MoSe_2$–WSe_2 (**g**), MoS_2–WS_2 (**h**) and alloyed $MoS_{0.64}Se_{1.36}$–$WS_{0.68}Se_{1.32}$ (**i**) lateral superlattices. The bandgaps of monolayer WSe_2, $MoSe_2$, WS_2, MoS_2, alloyed $MoS_{0.64}Se_{1.36}$ and $WS_{0.68}Se_{1.32}$ are 1.6 eV, 1.52 eV, 1.97 eV, 1.84 eV, 1.61 eV and 1.71 eV, respectively[46]. **j**, Schematic of an MOCVD-grown TMD–TMD lateral superlattice. **k**, Scanning electron microscope images of WS_2–WSe_2 lateral superlattices[47]. **l**, Schematic illustration of the lateral endoepitaxial growth of WS_2–WSe_2 MMHs. **m**–**p**, Photoluminescence mapping of monolayer WS_2–WSe_2 (**m** for WS_2 and **n** for WSe_2) and WS_2–MoS_2 (**o** for WS_2 and **p** for MoS_2) MMHs[48]. Panels reproduced with permission from: **a**–**e**, ref. 16, AAAS; **f**–**i**, ref. 46, Springer Nature Limited; **j**,**k**, ref. 47, AAAS; **l**–**p**, ref. 48, Springer Nature Limited.

in monolayer TMDs with atomically clean edges. These edges function as exclusive nucleation sites for the endoepitaxial growth of a second TMD, producing diverse lateral MMHs (such as WSe_2–WS_2, MoS_2–WS_2 and MoS_2–WSe_2) with atomically sharp heterojunction interfaces. Systematic microstructure and spectroscopic characterizations revealed periodic modulations of the chemical compositions, lattice strains and electronic bandgaps throughout the mosaic heterostructures (Fig. 2m–p). A patterned MoS_2–$MoSe_2$ lateral heterostructure array was also fabricated by replacing the Se atoms with S atoms in patterned $MoSe_2$ (ref. 49).

Synthesis of vertical heterostructures and moiré superlattices

Besides lateral integrated heterostructures and superlattices, large-scale vertically stacked van der Waals (vdW) heterostructures (vdWHs) and superlattices with high-quality interfaces have attracted considerable interest for tailoring electronic and optoelectronic properties and creating functional devices[50,51]. Although a wide range of vertical heterostructures have been prepared using exfoliation and restacking processes, developing a larger-scale synthetic approach to TMD-based vertical heterostructures would be an essential step in their potential integration in practical technologies.

One-step CVD synthesis of ReS_2/WS_2 vertical heterostructures with 100% overlap has been achieved using a designed substrate consisting of a single-crystalline Au (111) foil supported on a W–Re alloy foil[52]. This twinned growth can be attributed to selective epitaxy induced by the difference in adsorption energies between the different precursor molecules and the epitaxial substrate. A sequential two-step CVD growth method has also been used to prepare WSe_2/$MoSe_2$ vdWHs with sizes of up to 169 μm (ref. 50). CVD growth processes usually produce vdWHs with well-defined vdW epitaxial relationships and clean vdW interfaces, resulting in 0° or 60° (defined as AA or AB) stacks with strong interlayer coupling and efficient interlayer charge transfer[53,54]. Recent studies have suggested that substrate defects may play an important role in the unidirectional growth of TMDs on hexagonal boron nitride (hBN) substrates[55,56]. For example, a single W atom stably trapped on a hBN surface at a single vacancy site was directly observed using high-angle annular dark field-scanning transmission electron microscopy (HAADF-STEM), consistent with the defect-assisted nucleation model[55].

In general, the atomically smooth surface of 2D TMDs makes them excellent templates for producing large-area ultrathin 2D materials including metallic tellurides[57], semiconducting SnS_2 (ref. 51), ferromagnetic $CrSe_2$ (ref. 58) and non-layered materials (NiSe (ref. 59)) that are difficult to obtain on routine SiO_2/Si substrates and can provide high-quality vertical vdWHs. For example, under the same growth conditions, metallic $NbTe_2$ nanosheets grown on a WSe_2 substrate are uniform monolayers (~1.0 nm), whereas those on SiO_2/Si substrates show relatively broad thickness distributions ranging from a few to hundreds of nanometres. In the vdW epitaxy process, the atomically smooth, dangling-bond-free surface of WSe_2 can effectively weaken the interaction between adatoms and growth substrates. This ensures a lower energy barrier for surface diffusion, leading to a migration coefficient that is higher than those on SiO_2/Si substrates and is favourable for the rapid lateral growth of ultrathin 2D nanosheets.

In addition to solid-source thermal CVD, chalcogenization of transition metal or metal oxide thin films provides another way to prepare large-area vertical TMD heterostructures[60,61]. Specifically, the preparation usually includes the multi-step successive deposition of different metal films followed by one-step sulfurization/selenization[62,63]. However, thermal etching and interface alloying are inevitable during high-temperature sulfurization/selenization[64]. To this end, a layer-by-layer growth mode has recently been reported for the growth of multi-vertical vdWHs (Fig. 3a–c) by sulfurization (or selenization) of thin metal films[13]. In this design, multiple cycles of two-step growth (the deposition of one metal film followed by the sulfurization/selenization process) were consecutively conducted to prepare multi-vdWH films. To avoid thermal etching and alloy formation, the growth sequence must follow the growth temperatures of the 2DACs, strictly obeying a high-to-low temperature sequence. With this approach, the numbers of layers of 2DACs in the vdWHs can be precisely controlled, producing a series of double-block, triple-block, four-block and even five-block vdWH thin films.

Vertical stacking of two or more atomically thin layers with lattice mismatches or twist angles can lead to periodic supercells; that is, moiré superlattices[65]. Moiré superlattices provide a highly tunable strategy to tailor periodic potential modulations on a scale much larger than atomic lattices, which can lead to tunable charge correlation and exotic physical properties—as highlighted by the unconventional superconductivity observed in twisted bilayer magic-angle graphene[66]. Similarly, the periodic moiré superlattices formed in heterobilayer or twisted TMDs also play a key role in regulating the electronic band structure and producing a series of interesting physical phenomena[67,68]. Heterobilayer TMDs grown directly by the CVD method are typically not twisted, but thermodynamically favour 2H or 3R stacking[53,54]. Nonetheless, the wide range of TMDs with distinct lattice parameters can readily produce moiré superlattices with vastly different supercell sizes in epitaxially grown vdW heterobilayers (Fig. 3d–g), including SnS_2/WSe_2 (3.63 nm) (Fig. 3e)[69], $CrSe_2$/WSe_2 (3.63 nm) (Fig. 3f)[58] and VSe_2/WSe_2 (14.7 nm) (Fig. 3g)[70]. Interestingly, by growing WS_ySe_{2-y}/WS_xSe_{2-x} vdWHs with different S/Se ratios and controllable lattice mismatches, a recent study realized widely tunable moiré superlattices with supercell sizes from 10 to 45 nm (ref. 71).

There is a substantial energy barrier to forming twisted bilayer TMDs due to the rotations of the interlayers. To address this challenge, a recent study introduced the element tin (Sn) into the CVD system to reduce the interlayer rotation energy, producing twisted bilayer WS_2 and WSe_2 (refs. 72,73). Subsequently, a reconfiguring nucleation CVD strategy has been proposed for directly synthesizing twisted bilayer MoS_2 with a wide range of twist angles from 0° to 120° (Fig. 3h,i). Twist-angle-dependent moiré superlattices can be clearly observed, and the interlayer coupling shows a strong relationship to the twist angles (Fig. 3j). However, twisted growth by design[65] has not yet been realized for bilayer TMDs.

Supertwisted spiral growth provides another promising way to prepare twisted multilayer TMDs[74]. Screw dislocation is a line defect that shears part of the crystal lattice along a specific direction[75]. In layered materials, the shear typically occurs along the out-of-plane direction and connects layers into one continuous layer. When a screw dislocation spiral takes place on a flat substrate with Euclidean surfaces, a non-twisted triangular spiral shape with aligned edges is formed (Fig. 3k, left, and Fig. 3l). Interestingly, once a non-Euclidean surface with a curved geometrical space is adopted, supertwisted spirals of WS_2 can be produced (Fig. 3k, right, and Fig. 3m). Specifically, when a vdW crystal grows on a conical surface with a non-Euclidean geometry, it will bend following the curvature of the cone. For every period, the new edge will twist by an angle relative to the edges of the last period. Continuously twisted superstructures with a large and variable twist angle can also be realized by designing the shape of the non-Euclidean surface, as confirmed by the formation of moiré superlattices (Fig. 3n–o).

Synthesis of vertical superlattices

The direct CVD growth of higher-order vertical heterostructures or superlattices with three or more constituent blocks is a considerable challenge due to the need to establish a delicate balance between the growth of a new atomic layer and the thermal etching of the previous layers. To this end, the MOCVD process has shown interesting potential for the direct growth of vdW superlattices (vdWSLs) consisting of heteroepitaxially stacked with MoS_2, WS_2 and WSe_2 monolayers[76]. A series of cross-sectional STEM images taken at each growth step

https://doi.org/10.1038/s41563-024-01989-8

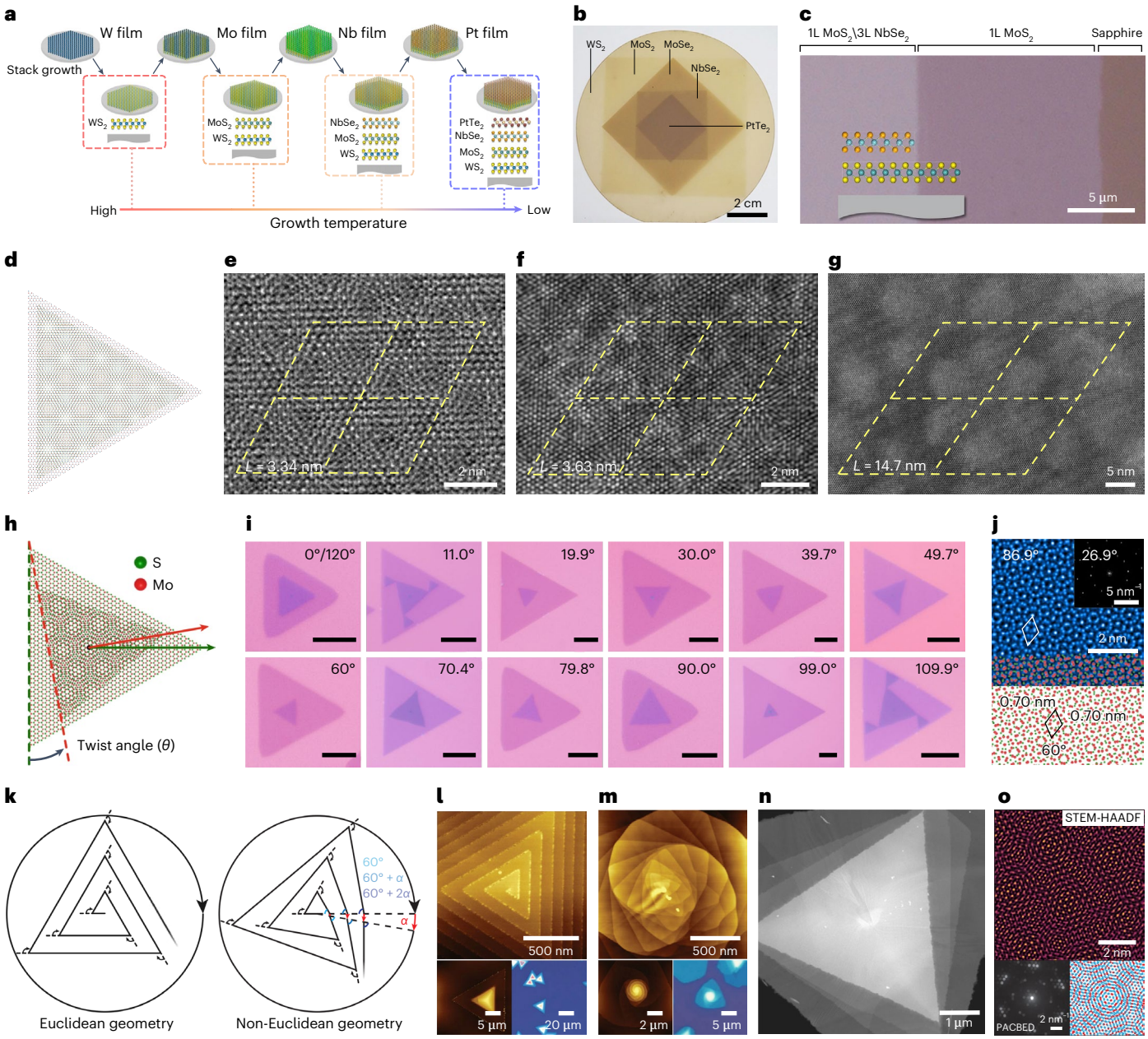

Fig. 3 | Synthesis of 2D vertical heterostructures and moiré superlattices. **a**, Schematic showing the stack growth of multiblock vdWHs guided by the high–low-temperature strategy. **b**, Optical microscope images of wafer-scale five-block vdWH ($PtTe_2/NbSe_2/MoSe_2/MoS_2/WS_2$). **c**, Optical microscope image of a stack-grown double-block vdWH on sapphire, consisting of monolayer (1L) MoS_2 at the bottom and 3L $NbSe_2$ at the top[13]. The inset shows the corresponding atomic model diagram. **d**, Schematic of a moiré superlattice resulting from the stacking of lattice-mismatched layers. **e**–**g**, Moiré superlattices produced from epitaxially grown vdW heterobilayers of SnS_2/WSe_2 (3.34 nm; **e**)[69], $CrSe_2/WSe_2$ (3.63 nm; **f**)[58] and VSe_2/WSe_2 (14.7 nm; **g**). The yellow dashed lines in **e**–**g** represent the 2 × 2 moiré period, in which the values for one moiré period (L) are 3.4 nm, 3.63 nm and 14.7 nm, respectively[70]. **h**, Schematic atomic configuration of twisted bilayer (TB)-MoS_2. **i**, Optical microscope images of the as-grown TB-MoS_2 with twist angles from 0° to 120°. Scale bars, 10 μm. **j**, Atomic-resolution HAADF-STEM image of a commensurate superlattice (top) and corresponding atom models of 86.9°-TB-MoS_2 (bottom). The inset is the corresponding fast Fourier transform patterns. The white rhombus indicates a moiré period of 0.7 nm (ref. 96). **k**, Schematic diagram of Euclidean and non-Euclidean geometries. Left: for a triangular spiral on a Euclidean surface, at each vertex the edge direction changes 120°; thus, after three vertices, it turns 360°, which makes the edge parallel to its original orientation. Right: for a triangular spiral on an unfolded non-Euclidean conical surface, after every three vertices, the edge goes across the dashed line once, so it needs to turn an extra angle to keep itself straight. **l**, Atomic force microscope (top and bottom left) and optical microscope (bottom right) images of aligned WS_2 spirals grown on a flat SiO_2/Si substrate. **m**, Atomic force microscope (top and bottom left) and optical microscope (bottom right) images of representative WS_2 supertwisted spirals with a twist angle grown around WO_x particles on SiO_2/Si substrates. **n**, HAADF image of a supertwisted WS_2 plate with edges of different layers highlighted by different shades of grey. **o**, High-resolution HAADF and selected area electron diffraction images collected from the three-layer area in **n**[74]. PACBED, position-averaged convergent beam electron diffraction. Panels reproduced with permission from: **a**–**c**, ref. 13, Springer Nature Limited; **e**, ref. 69, Springer Nature Limited; **f**, ref. 58, Springer Nature Limited; **g**, ref. 70, Springer Nature Limited; **h**–**j**, ref. 96 under a Creative Commons licence CC BY 4.0; **k**–**o**, ref. 74, AAAS.

https://doi.org/10.1038/s41563-024-01989-8

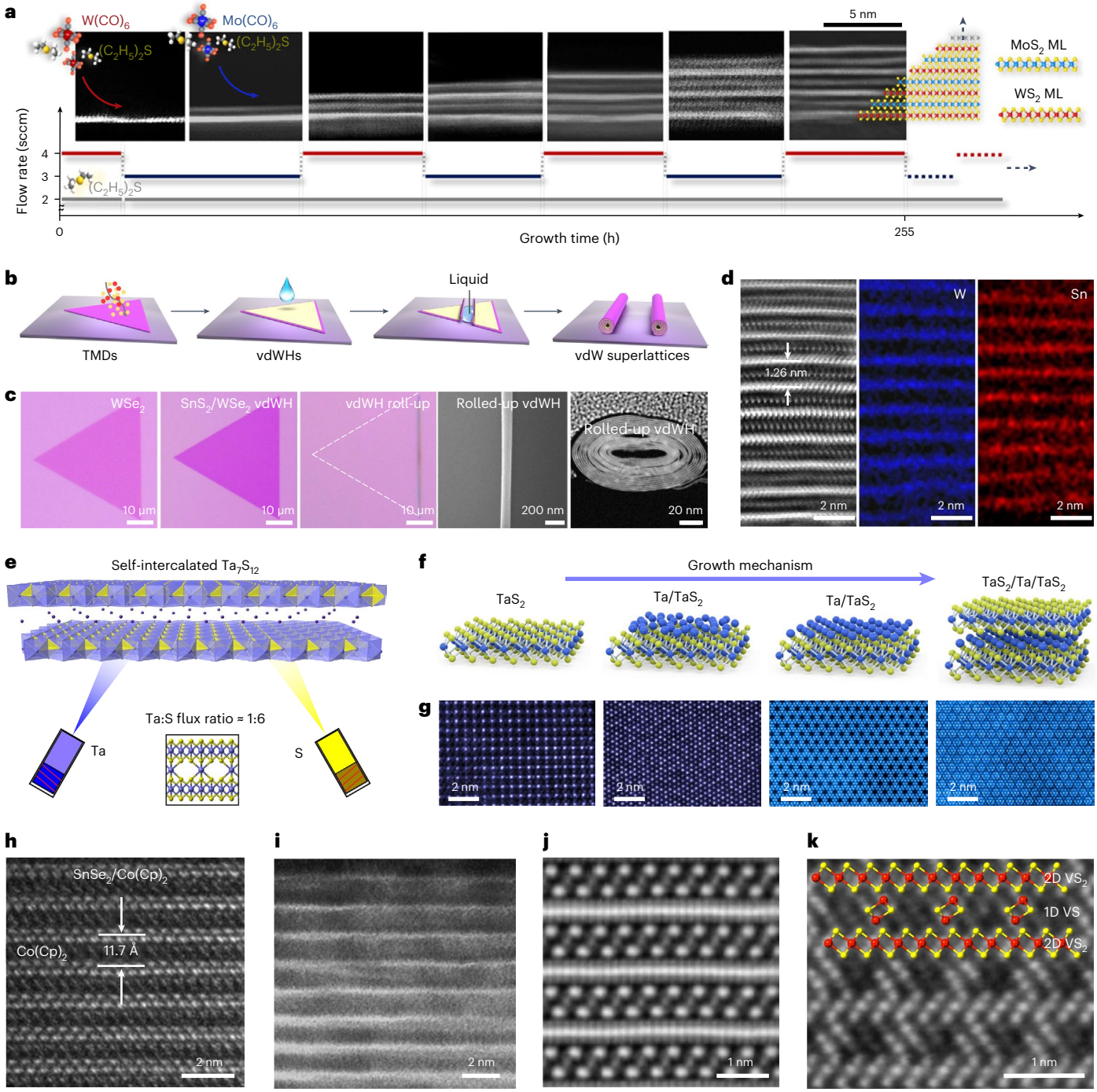

Fig. 4 | Two-dimensional vertical superlattices. a, Top: MOCVD-grown MoS_2/WS_2 superlattices (right) and a series of cross-sectional HAADF-STEM images of the MoS_2/WS_2 supperlattices from 1 to 7 ML stacks (left). Bottom: flow-rate modulations of MO precursors for the growth of MoS_2/WS_2 superlattices[76]. **b**,**c**, Schematic illustration of the fabrication process of rolled-up vdW superlattices. **d**, Structural characterizations of rolled-up SnS_2/WSe_2 vdWSLs, including the cross-sectional STEM image (left) and the corresponding energy disperse spectroscopy for W (centre) and Sn (right)[69]. **e**, Schematic showing the growth of self-intercalated Ta_7S_{12} by molecular beam epitaxy in a high-Ta-flux environment. **f**, Schematic depicting the layer-by-layer growth of 2D crystal integrated circuits. **g**, Atomic-resolution STEM images of 2D crystals of (from left to right) 25% Ta-intercalated Ta_9S_{16}, 50% Ta-intercalated $Ta_{10}S_{16}$, 66.7% Ta-intercalated Ta_8Se_{12} and 100% Ta-intercalated Ta_9Se_{12} (ref. 79). **h**, HAADF-STEM image of a $SnSe_2/Co(Cp)_2$ superlattice nanoflake[81]. **i**, High-resolution cross-sectional STEM image of a TaS_2/MBA superlattice. The bright regions correspond to the H-TaS_2 monolayers[78]. **j**, HAADF-STEM image of a NbS_2/Ba_3NbS_5 superlattice[82]. **k**, Cross-section ADF image of a 2D VS_2–1D VS superlattice[85]. Panels reproduced with permission from: **a**, ref. 76, Springer Nature Limited; **b**–**d**, ref. 69, Springer Nature Limited; **e**–**g**, ref. 79, Springer Nature Limited; **h**, ref. 81, American Chemical Society; **i**, ref. 78, Springer Nature Limited; **j**, ref. 82, AAAS; **k**, ref. 85, Springer Nature Limited.

demonstrate the precise monolayer-by-monolayer stacking of WS_2 and MoS_2 (Fig. 4a). However, at present the MOCVD method is limited to the MX_2 (M = Mo, W; X = S, Se) system due to the limited availability of metal–organic precursors.

Instead of sequential layer-by-layer growth, rolling up large-area vdWHs offers an alternative approach to producing high-order vdWSLs with a larger number of alternating units[69] (Fig. 4b,c). For example, when CVD-grown SnS_2/WSe_2 vdWHs are exposed to an

https://doi.org/10.1038/s41563-024-01989-8

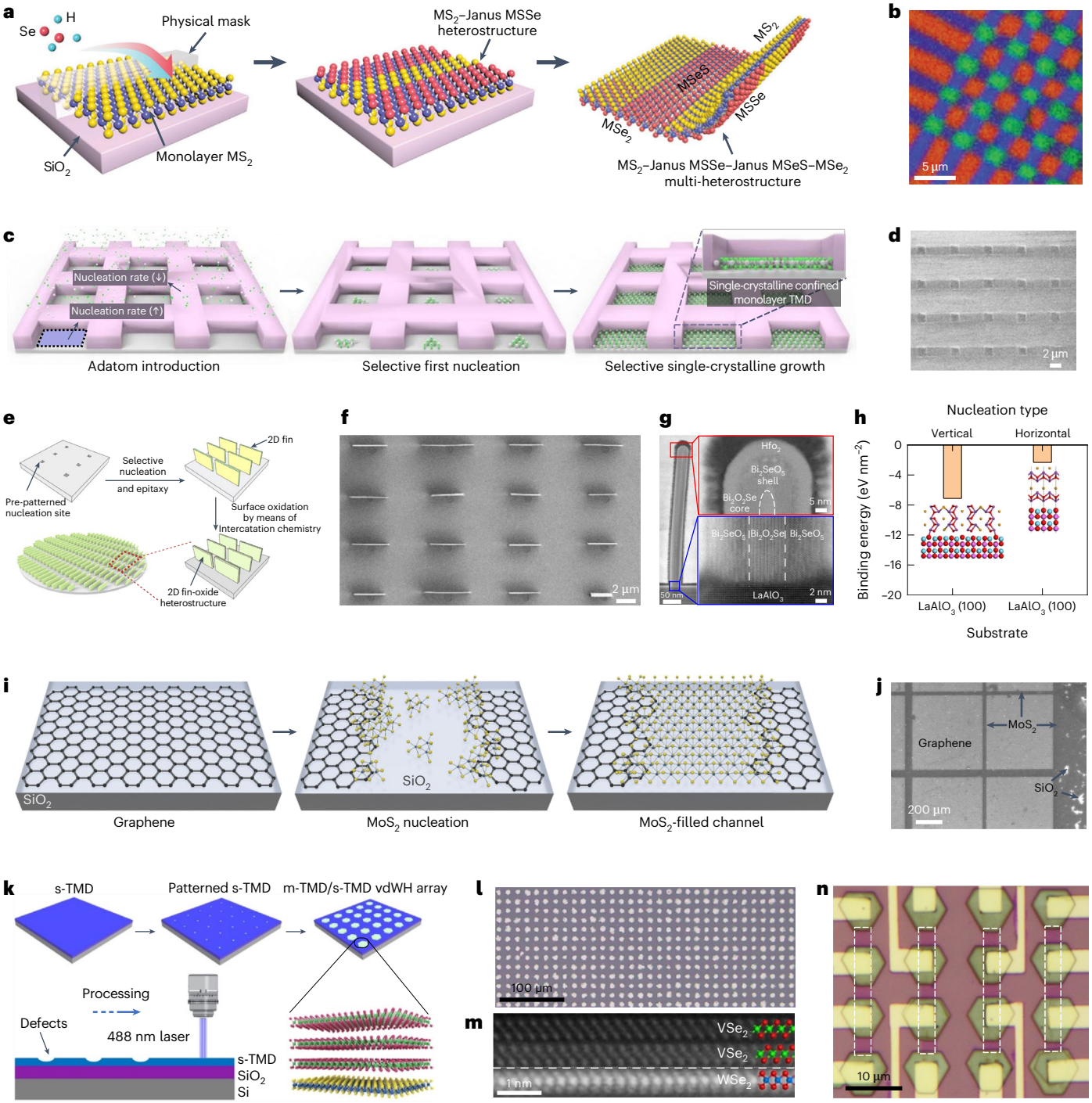

Fig. 5 | Site-controlled growth. a,**b**, Site-selective growth by lithographic patterning. **a**, Schematic representation of the lithographic-patterning-assisted atomic-layer substitution process. M is W or Mo. **b**, Spatially resolved Raman mapping of the A_{1g} mode intensity of an MoS_2–Janus MoSSe–Janus MoSeS–$MoSe_2$ MMH. The red, blue and green areas represent the MoS_2, Janus and $MoSe_2$ regions, respectively[89]. **c**,**d**, Site-selective growth by geometric confinement. **c**, Schematic of the process of growth by geometric confinement. **d**, SEM image of single-domain WSe_2 arrays selectively synthesized in 2 μm sapphire pockets fabricated with a-SiO_2 trenches[90]. **e**–**h**, Direct growth on fin way, vertically aligned arrays of 2D fin-oxide heterostructures with the epitaxial direction perpendicular to the substrate. **e**, Schematic of wafer-scale 2D fin–oxide core–shell heterostructure arrays. **f**, Typical SEM image of a periodic array of 2D fin–oxide core–shell heterostructures. **g**, Cross-sectional STEM images of a 2D Bi_2O_2Se–Bi_2SeO_5 fin–oxide heterostructure covered with a HfO_2 dielectric layer. **h**, DFT calculations of the binding energies of a Bi_2O_2Se island with different nucleation types on the $LaAlO_3$ (100) surfaces[91]. **i**,**j**, Direct growth between patterned metal electrodes. **i**, Schematic of the MoS_2–graphene heterostructure growth process. **j**, SEM image of the chemically grown MoS_2 between the edges of the graphene[92]. **k**–**n**, Laser-induced vdW epitaxy at controlled locations. **k**, Schematic of the defects array fabricated through laser cauterization and defect-induced growth of an s-TMD/m-TMD vdWH array. **l**, Optical microscope image of large-area VSe_2/MoS_2 vdWH arrays. **m**, Cross-sectional STEM image of a VSe_2/WSe_2 vdWH with the epitaxial direction parallel to the substrate. **n**, Optical microscope image of back-gated WSe_2 transistors with neighbouring VSe_2 nanoplates acting as the synthetic vdW contacts[70]. Figure reproduced with permission from: **a**,**b**, ref. 89, National Academy of Sciences; **c**,**d**, ref. 90, Springer Nature Limited; **e**–**h**, ref. 91, Springer Nature Limited; **i**,**j**, ref. 92, Springer Nature Limited; **k**–**n**, ref. 70, Springer Nature Limited.

ethanol–water–ammonia solution, the intercalation of solvent molecules between the 2D heterostructures and the substrate delaminates the vdWHs from the substrate and drives a spontaneous rolling-up process to form rolled-up structures consisting of alternating layers of monolayer WSe_2 and monolayer SnS_2 (Fig. 4d). The distance between the monolayer SnS_2 and the monolayer WSe_2 in the vdWSL (0.63 nm) is essentially the same as the intrinsic interlayer distance, indicating a high-quality vdW interface with little interlayer contamination. This rolling-up strategy can be extended to produce various 2D/2D vdWSLs and more complex three-component 2D/2D/2D vdWSLs, as well as multi-dimensional vdWSLs consisting of a mixture of 2D with 3D or 1D materials. The resulting high-order vdWSLs with variable dimensions and compositions feature highly tunable band alignments and chiralities, providing an intriguing material platform for both fundamental research and technological applications.

Another strategy for realizing high-order 2D superlattices involves the intercalation of selected atomic or molecular species into layered crystals[77–80]. For example, the self-intercalation of native atoms into layered TMDs can be directly achieved by CVD or molecular beam epitaxy[79] (Fig. 4e). A metal-rich chemical potential provides the driving force for the self-intercalation of a metal into layered materials. By varying the ratio of intercalating atoms, Ta_xS_y (non-stoichiometric with respect to TaS_2) films with different concentrations of intercalated Ta can be grown, generating a series of well-defined superstructures that have been verified by ADF-STEM imaging (Fig. 4f,g). The chemical or electrochemical intercalation of self-assembled molecular layers within layered TMDs offers another versatile strategy for producing high-order superlattices with tailored interlayer coupling (MoS_2/cetyl-trimethylammonium bromide (CTAB)[77], as a unique bulk material with some similar physical properties to monolayer materials) and structural topologies that may induce exotic properties such as modulated superconductivity in $SnSe_2/Co(Cp)_2$ superlattices (Fig. 4h)[81] and chirality-induced spin selectivity in TaS_2/chiral-methylbenzylamine (MBA) superlattices[78] (Fig. 4i).

Natural vdW superlattices that also form between alternating TMD layers and commensurate blocking layers provide an effective pathway to tailoring the dimensionality and electronic properties of layered materials (for example, NbS_2/Ba_3NbS_5 (Fig. 4j)[82], $(SmS)(TaS_2)$ (ref. 83) and $(PbSe)_{1.14}/NbSe_2$ (ref. 84)). Here the blocking layer can weaken or decouple the interlayer coupling between neighbouring active TMD monolayers to produce a distinct family of 2D superconductors (Fig. 4j) and topological insulators. Furthermore, an intrinsic heterodimensional superlattice consisting of alternating layers of 2D VS_2 and a 1D VS chain array (Fig. 4k) has been realized with a one-step molten-salt CVD method[85], providing an intriguing materials system for uncovering new physical properties and developing new functional devices.

Site-selective growth

The selective growth of 2D materials at specifically designated locations or pre-patterned surfaces provides a promising route to scalable integration without having to form single crystals on a wafer scale. However, most synthetic approaches to 2D crystals rely on random nucleation, with relatively limited control over the spatial location. Robust and scalable synthesis of 2DACs and their heterostructure arrays with precisely controlled spatial locations and periodic arrangements represents an important research direction in the synthesis of 2D materials.

Lithographic patterning of the growth substrate to create site-selective nucleation offers a rational strategy for site-selective growth of 2D crystals. Seeded growth is one of the most comonly used strategies for producing 2DAC arrays. For example, precursors pre-seeded at designated locations on the growth substrate allow in situ feeding and reactions, and thus the formation of periodic 2DAC arrays. Examples include carbon-containing seeds (polymethyl methacrylate, PMMA) for graphene arrays[86] and molybdenum-containing seed particles (ammonium heptamolybdate or MoO_3) for MoS_2 arrays[87]. An additive manufacturing method provides another feasible route to growing arbitrarily patterned monolayer MoS_2 (ref. 88). In this method, O_2-plasma-treated SiO_2/Si substrate modification in combination with a salt growth promoter (perylene-3,4,9,10-tetracarboxylic acid tetrapotassium) plays a crucial role in site-controlled growth. A remote hydrogen-plasma-assisted atomic-layer substitution[89] provides a powerful approach for synthesizing Janus monolayer TMDs at room temperature (Fig. 5a). This strategy can be combined with lithographic pattering that exposes selective regions for Janus conversion to enable the production of artificial in-plane multi-heterostructures with specific spatial compositions, such as monolayer MS_2–Janus MSSe, MoS_2–Janus MoSSe–Janus MoSeS–$MoSe_2$ (Fig. 5b).

A deterministic confined-growth technique was recently reported to realize non-epitaxial growth of wafer-scale single-domain 2D monolayer arrays and vertical growth of bilayers (both homo- and heterogeneously) on arbitrary substrates[90] (Fig. 5c,d). In this approach, the selective growth area was defined by pattering SiO_2 masks. The nucleation selectively occurred in the patterned c-Al_2O_3 or a-HfO_2 surface instead of the a-SiO_2 due to the stronger binding interaction between WSe_2 and c-Al_2O_3/a-HfO_2. Once the patterned c-Al_2O_3/a-HfO_2 area was small enough, a single-domain monolayer nucleation event was allowed to ensure single-crystal domain growth at the designated patterning area. Further growth of WSe_2 or other TMDs with the same principle yields single-domain bilayer WSe_2 or MoS_2/WSe_2 heterostructures.

In addition to planar architectures, the direct growth of vertical-architecture 2DAC arrays holds important interest for integration with 3D Si fin field effect transistors (FinFETs) and the development of ultra-scaled transistors. A recent study reported wafer-scale, site-specific and high-density growth of a vertically unidirectionally aligned array of 2D layered Bi_2O_2Se fins (Fig. 5e–h)[91]. By using an abnormal insulating oxide substrate with a low-symmetry (such as $LaAlO_3$ (110) and MgO (110)), the vertical 2D Bi_2O_2Se showed a much lower binding energy than the flat one, whose nucleation is thermodynamically favourable (Fig. 5h). Circular pits with depths of several nanometres were fabricated by selective Ar-based ion-beam etching on the inch-scale MgO substrate to serve as the pre-patterned nucleation sites. Subsequently, an in situ ultraviolet (UV)-assisted oxidation process was developed to partially oxidize 2D Bi_2O_2Se fins into 2D Bi_2O_2Se–Bi_2SeO_5 fin–oxide core–shell heterostructure arrays (Fig. 5e–g). Such 2D fin–oxide epitaxial heterostructures with atomically flat and high-quality interfaces enable improved device performance and reliability of 2D FinFETs, opening up new avenues for the further extension of Moore's law.

The direct growth of 2DAC arrays between patterned metal electrodes offers an interesting strategy for scalable integration[92–94]. For example, pre-patterned graphene electrodes with dangling bonds at the edges offer effective nucleation points for the lateral growth of monolayer MoS_2, bridging the graphene electrode pairs (Fig. 5i) and allowing the scalable integration of high-performance devices and circuits based on MoS_2–graphene heterostructures[92] (Fig. 5j). Alternatively, metal-guided selective growth allows local control of the transition-metal-precursor vapour pressure, enabling the precise growth of TMD materials, as well as their lateral or vertical heterojunctions at desired locations over the entire wafer. This location selectivity means that metal-guided selective growth enables the synthetic integration of p-type WSe_2 and n-type $MoSe_2$ transistors and complementary metal–oxide–semiconductor inverters on the wafer scale[93].

Alternatively, site-selective vdW epitaxial growth of high-quality single-crystal metal TMDs (m-TMDs) on large-area 2D semiconductor TMDs (s-TMDs) offers a direct synthetic pathway to high-quality synthetic vdW contacts for the scalable integration of high-performance 2D FETs. To this end, a laser cauterization technology was used to selectively pattern nucleation sites on s-TMDs, at which m-TMDs may preferentially nucleate and grow to form periodic m-TMD/s-TMD vdWH

arrays[70] (Fig. 5k). In contrast to the traditional lithography technique, the direct laser cauterization patterning approach avoids the contamination of impurities. This allows local defects to be created at specific locations, minimizing undesired random nucleation in subsequent growth and enabling the highly selective growth of large m-TMD/s-TMD vdWH arrays with the yield up to ~99% (Fig. 5l). The cross-sectional STEM image in Fig. 5m clearly demonstrates that the synthesized vdWHs have near-ideal, atomically sharp and clean interfaces. This approach allows the direct synthetic integration of a high-performance FET array (Fig. 5n). Benefiting from high-quality vdW interfaces, the synthetic vdW contacted devices show a unity device yield and greatly improved performance over deposited metal contacts, achieving a current density beyond 1.7 mA μm^{-1} in bilayer WSe_2 transistors[95]. This general approach has been extended to realize a series of 2D vdWH arrays with widely varying chemical compositions, moiré superlattice structures and electronic properties.

Conclusion

The controllability, scalability and uniformity of CVD synthesis hold the key to advancing fully 2D ACs and their heterostructures for practical technologies. Efforts to synthesize 2DACs have been directed towards achieving controlled thicknesses, lateral sizes, interlayer coupling, heterostructure interfaces, crystalline phases and spatial locations. This Review highlights recent progress in the synthetic control of large-area single crystals, 2D lateral and vertical heterostructures, superlattices and site-controlled growth. Despite significant advancements, substantial challenges remain, such as the wafer-scale growth of heterostructures and superlattices with large-area uniformity, site-selective growth of periodic arrays of heterostructures on arbitrary substrates, homo-/heterobilayers with controlled twist angles and the synthesis of high-quality 2D heterostructures at the low temperatures that are compatible with state-of-the-art integrated circuit fabrication processes. Achieving these goals will require a thorough understanding and precise control of nucleation/growth processes. The interdisciplinary nature of this research involves collaboration between material scientists, chemists and engineers to push the boundaries of CVD synthesis and unlock the full potential of 2DACs for scalable technologies. Ultimately, overcoming these challenges is critical to design quantum photonic devices, explore new quantum phenomena in moiré materials (such as Wigner crystal states and Mott insulators) and realize the envisioned applications of 2DACs in next-generation electronics and optoelectronics.

References

1. Cao, W. et al. The future transistors. *Nature* **620**, 501–515 (2023).
2. Wang, S., Liu, X. & Zhou, P. The road for 2D semiconductors in the silicon age. *Adv. Mater.* **34**, 2106886 (2022).
3. *International Roadmap for Devices and Systems 2020 Edition* (IEEE, 2020); https://irds.ieee.org/editions/2020
4. Hills, G. et al. Modern microprocessor built from complementary carbon nanotube transistors. *Nature* **572**, 595–602 (2019).
5. Das, S. et al. Transistors based on two-dimensional materials for future integrated circuits. *Nat. Electron.* **4**, 786–799 (2021).
6. Zhu, K. et al. The development of integrated circuits based on two-dimensional materials. *Nat. Electron.* **4**, 775–785 (2021).
7. Chhowalla, M., Jena, D. & Zhang, H. Two-dimensional semiconductors for transistors. *Nat. Rev. Mater.* **1**, 16052 (2016).
8. Wu, F. et al. Vertical MoS_2 transistors with sub-1-nm gate lengths. *Nature* **603**, 259–264 (2022).
9. Liu, X. & Hersam, M. C. 2D materials for quantum information science. *Nat. Rev. Mater.* **4**, 669–684 (2019).
10. Li, J. et al. Synthesis of ultrathin metallic MTe_2(M = V, Nb, Ta) single-crystalline nanoplates. *Adv. Mater.* **30**, 1801043 (2018).
11. Liu, Y. et al. Van der Waals heterostructures and devices. *Nat. Rev. Mater.* **1**, 16042 (2016).
12. Andrei, E. Y. et al. The marvels of moiré materials. *Nat. Rev. Mater.* **6**, 201–206 (2021).
13. Zhou, Z. J. et al. Stack growth of wafer-scale van der Waals superconductor heterostructures. *Nature* **621**, 499–505 (2023).
14. Zhang, C. et al. Interlayer couplings, moiré patterns, and 2D electronic superlattices in MoS_2/WSe_2 hetero-bilayers. *Sci. Adv.* **3**, e1601459 (2017).
15. Zhang, T., Wang, J., Wu, P., Lu, A.-Y. & Kong, J. Vapour-phase deposition of two-dimensional layered chalcogenides. *Nat. Rev. Mater.* **8**, 799–821 (2023).
16. Zhang, Z. W. et al. Robust epitaxial growth of two-dimensional heterostructures, multiheterostructures, and superlattices. *Science* **357**, 788–792 (2017).
17. Zuo, Y. G. et al. Robust growth of two-dimensional metal dichalcogenides and their alloys by active chalcogen monomer supply. *Nat. Commun.* **13**, 1007 (2022).
18. Lin, Y.-C. et al. Wafer-scale MoS_2 thin layers prepared by MoO_3 sulfurization. *Nanoscale* **4**, 6637–6641 (2012).
19. Zhan, Y. J., Liu, Z., Najmaei, S., Ajayan, P. M. & Lou, J. Large-area vapor-phase growth and characterization of MoS_2 atomic layers on a SiO_2 substrate. *Small* **8**, 966–971 (2012).
20. Lin, H. et al. Growth of environmentally stable transition metal selenide films. *Nat. Mater.* **18**, 602–607 (2019).
21. Zhang, Z. et al. Ultrafast growth of large single crystals of monolayer WS_2 and WSe_2. *Natl Sci. Rev.* **7**, 737–744 (2020).
22. Zhang, Z. et al. Highly selective synthesis of monolayer or bilayer WSe_2 single crystals by pre-annealing the solid precursor. *Chem. Mater.* **33**, 1307–1313 (2021).
23. Lee, Y. H. et al. Synthesis of large-area MoS_2 atomic layers with chemical vapor deposition. *Adv. Mater.* **24**, 2320–2325 (2012).
24. Kang, K. et al. High-mobility three-atom-thick semiconducting films with wafer-scale homogeneity. *Nature* **520**, 656–660 (2015).
25. Zhu, J. et al. Low-thermal-budget synthesis of monolayer molybdenum disulfide for silicon back-end-of-line integration on a 200 mm platform. *Nat. Nanotechnol.* **18**, 456–463 (2023).
26. Wang, Q. et al. Layer-by-layer epitaxy of multi-layer MoS_2 wafers. *Natl Sci. Rev.* **9**, nwac077 (2022).
27. Yang, P. et al. Batch production of 6-inch uniform monolayer molybdenum disulfide catalyzed by sodium in glass. *Nat. Commun.* **9**, 979 (2018).
28. Xia, Y. et al. 12-inch growth of uniform MoS_2 monolayer for integrated circuit manufacture. *Nat. Mater.* **22**, 1324–1331 (2023).
29. Wang, Q. et al. Wafer-scale highly oriented monolayer MoS_2 with large domain sizes. *Nano Lett.* **20**, 7193–7199 (2020).
30. Xue, G. et al. Modularized batch production of 12-inch transition metal dichalcogenides by local element supply. *Sci. Bull.* **68**, 1514–1521 (2023).
31. Chen, J. et al. Chemical vapor deposition of large-size monolayer $MoSe_2$ crystals on molten glass. *J. Am. Chem. Soc.* **139**, 1073–1076 (2017).
32. Xu, X. L. et al. Seeded 2D epitaxy of large-area single-crystal films of the van der Waals semiconductor 2H $MoTe_2$. *Science* **372**, 195–200 (2021).
33. Li, T. T. et al. Epitaxial growth of wafer-scale molybdenum disulfide semiconductor single crystals on sapphire. *Nat. Nanotechnol.* **16**, 1201–1207 (2021).
34. Dong, J., Zhang, L., Dai, X. & Ding, F. The epitaxy of 2D materials growth. *Nat. Commun.* **11**, 5862 (2020).
35. Wang, J. H. et al. Dual-coupling-guided epitaxial growth of wafer-scale single-crystal WS_2 monolayer on vicinal *a*-plane sapphire. *Nat. Nanotechnol.* **17**, 33–38 (2022).
36. Fu, J.-H. et al. Oriented lateral growth of two-dimensional materials on *c*-plane sapphire. *Nat. Nanotechnol.* **18**, 1289–1294 (2023).

37. Yang, P. et al. Epitaxial growth of inch-scale single-crystal transition metal dichalcogenides through the patching of unidirectionally orientated ribbons. *Nat. Commun.* **13**, 3238 (2022).
38. Yang, P. et al. Epitaxial growth of centimeter-scale single-crystal MoS_2 monolayer on Au(111). *ACS Nano* **14**, 5036–5045 (2020).
39. Zheng, P. et al. Universal epitaxy of non-centrosymmetric two-dimensional single-crystal metal dichalcogenides. *Nat. Commun.* **14**, 592 (2023).
40. Son, Y. et al. Observation of switchable photoresponse of a monolayer WSe_2–MoS_2 lateral heterostructure via photocurrent spectral atomic force microscopic imaging. *Nano Lett.* **16**, 3571–3577 (2016).
41. Yoo, Y., Degregorio, Z. P. & Johns, J. E. Seed crystal homogeneity controls lateral and vertical heteroepitaxy of monolayer MoS_2 and WS_2. *J. Am. Chem. Soc.* **137**, 14281–14287 (2015).
42. Duan, X. et al. Lateral epitaxial growth of two-dimensional layered semiconductor heterojunctions. *Nat. Nanotechnol.* **9**, 1024–1030 (2014).
43. Huang, C. et al. Lateral heterojunctions within monolayer $MoSe_2$–WSe_2 semiconductors. *Nat. Mater.* **13**, 1096–1101 (2014).
44. Gong, Y. J. et al. Vertical and in-plane heterostructures from WS_2/MoS_2 monolayers. *Nat. Mater.* **13**, 1135–1142 (2014).
45. Li, M.-Y. et al. Epitaxial growth of a monolayer WSe_2-MoS_2 lateral p-n junction with an atomically sharp interface. *Science* **349**, 524–528 (2015).
46. Sahoo, P. K., Memaran, S., Xin, Y., Balicas, L. & Gutierrez, H. R. One-pot growth of two-dimensional lateral heterostructures via sequential edge-epitaxy. *Nature* **553**, 63–67 (2018).
47. Xie, S. et al. Coherent, atomically thin transition-metal dichalcogenide superlattices with engineered strain. *Science* **359**, 1131–1136 (2018).
48. Zhang, Z. W. et al. Endoepitaxial growth of monolayer mosaic heterostructures. *Nat. Nanotechnol.* **17**, 493–499 (2022).
49. Mahjouri-Samani, M. et al. Patterned arrays of lateral heterojunctions within monolayer two-dimensional semiconductors. *Nat. Commun.* **6**, 7749 (2015).
50. Gong, Y. et al. Two-step growth of two-dimensional WSe_2/$MoSe_2$ heterostructures. *Nano Lett.* **15**, 6135–6141 (2015).
51. Yang, T. et al. Van der Waals epitaxial growth and optoelectronics of large-scale WSe_2/SnS_2 vertical bilayer p–n junctions. *Nat. Commun.* **8**, 1906 (2017).
52. Zhang, T. et al. Twinned growth behaviour of two-dimensional materials. *Nat. Commun.* **7**, 13911 (2016).
53. Zhang, Q. et al. Two-dimensional layered heterostructures synthesized from core–shell nanowires. *Angew. Chem. Int. Ed.* **54**, 8957–8960 (2015).
54. Zhang, J. et al. Observation of strong interlayer coupling in MoS_2/WS_2 heterostructures. *Adv. Mater.* **28**, 1950–1956 (2016).
55. Zhang, F. et al. Full orientation control of epitaxial MoS_2 on hBN assisted by substrate defects. *Phys. Rev. B* **99**, 155430 (2019).
56. Zhang, X. T. et al. Defect-controlled nucleation and orientation of WSe_2 on hBN: a route to single-crystal epitaxial monolayers. *ACS Nano* **13**, 3341–3352 (2019).
57. Wu, R. et al. Van der Waals epitaxial growth of atomically thin 2D metals on dangling-bond-free WSe_2 and WS_2. *Adv. Funct. Mater.* **29**, 1806611 (2019).
58. Li, B. et al. Van der Waals epitaxial growth of air-stable $CrSe_2$ nanosheets with thickness-tunable magnetic order. *Nat. Mater.* **20**, 818–825 (2021).
59. Zhao, B. et al. Van der Waals epitaxial growth of ultrathin metallic NiSe nanosheets on WSe_2 as high performance contacts for WSe_2 transistors. *Nano Res.* **12**, 1683–1689 (2019).
60. Seok, H. et al. Low-temperature synthesis of wafer-scale MoS_2–WS_2 vertical heterostructures by single-step penetrative plasma sulfurization. *ACS Nano* **15**, 707–718 (2021).
61. Choudhary, N. et al. Centimeter scale patterned growth of vertically stacked few layer only 2D MoS_2/WS_2 van der Waals heterostructure. *Sci. Rep.* **6**, 25456 (2016).
62. Xue, Y. et al. Scalable production of a few-layer MoS_2/WS_2 vertical heterojunction array and its application for photodetectors. *ACS Nano* **10**, 573–580 (2016).
63. Yuan, J. et al. Wafer-scale fabrication of two-dimensional PtS_2/$PtSe_2$ heterojunctions for efficient and broad band photodetection. *ACS Appl. Mater. Interfaces* **10**, 40614–40622 (2018).
64. Wu, C.-R. et al. Establishment of 2D crystal heterostructures by sulfurization of sequential transition metal depositions: preparation, characterization, and selective growth. *Nano Lett.* **16**, 7093–7097 (2016).
65. Liu, C. et al. Designed growth of large bilayer graphene with arbitrary twist angles. *Nat. Mater.* **21**, 1263–1268 (2022).
66. Cao, Y. et al. Unconventional superconductivity in magic-angle graphene superlattices. *Nature* **556**, 43–50 (2018).
67. Susarla, S. et al. Hyperspectral imaging of exciton confinement within a moiré unit cell with a subnanometer electron probe. *Science* **378**, 1235–1239 (2022).
68. Devakul, T., Crépel, V., Zhang, Y. & Fu, L. Magic in twisted transition metal dichalcogenide bilayers. *Nat. Commun.* **12**, 6730 (2021).
69. Zhao, B. et al. High-order superlattices by rolling up van der Waals heterostructures. *Nature* **591**, 385–390 (2021).
70. Li, J. et al. General synthesis of two-dimensional van der Waals heterostructure arrays. *Nature* **579**, 368–374 (2020).
71. Fortin-Deschênes, M., Watanabe, K., Taniguchi, T. & Xia, F. Van der Waals epitaxy of tunable moirés enabled by alloying. *Nat. Mater.* **23**, 339–346 (2024).
72. Shao, G. et al. Twist angle-dependent optical responses in controllably grown WS_2 vertical homojunctions. *Chem. Mater.* **32**, 9721–9729 (2020).
73. Zheng, H. et al. Strong interlayer coupling in twisted transition metal dichalcogenide moiré superlattices. *Adv. Mater.* **35**, 2210909 (2023).
74. Zhao, Y. Z. et al. Supertwisted spirals of layered materials enabled by growth on non-Euclidean surfaces. *Science* **370**, 442–445 (2020).
75. Wang, Z.-J. et al. Conversion of chirality to twisting via sequential one-dimensional and two-dimensional growth of graphene spirals. *Nat. Mater.* **23**, 331–338 (2024).
76. Jin, G. et al. Heteroepitaxial van der Waals semiconductor superlattices. *Nat. Nanotechnol.* **16**, 1092–1098 (2021).
77. Wang, C. et al. Monolayer atomic crystal molecular superlattices. *Nature* **555**, 231–236 (2018).
78. Qian, Q. et al. Chiral molecular intercalation superlattices. *Nature* **606**, 902–908 (2022).
79. Zhao, X. et al. Engineering covalently bonded 2D layered materials by self-intercalation. *Nature* **581**, 171–177 (2020).
80. Zhang, L. et al. 2D atomic crystal molecular superlattices by soft plasma intercalation. *Nat. Commun.* **11**, 5960 (2020).
81. Li, Z. et al. Molecule-confined engineering toward superconductivity and ferromagnetism in two-dimensional superlattice. *J. Am. Chem. Soc.* **139**, 16398–16404 (2017).
82. Devarakonda, A. et al. Clean 2D superconductivity in a bulk van der Waals superlattice. *Science* **370**, 231–236 (2020).
83. Dolotko, O. et al. Unprecedented generation of 3D heterostructures by mechanochemical disassembly and re-ordering of incommensurate metal chalcogenides. *Nat. Commun.* **11**, 3005 (2020).
84. Zhong, H. et al. Revealing the two-dimensional electronic structure and anisotropic superconductivity in a natural van der Waals superlattice $(PbSe)_{1.14}NbSe_2$. *Phys. Rev. Mater.* **7**, L041801 (2023).

85. Zhou, J. et al. Heterodimensional superlattice with in-plane anomalous Hall effect. *Nature* **609**, 46–51 (2022).
86. Wu, W. et al. Growth of single crystal graphene arrays by locally controlling nucleation on polycrystalline Cu using chemical vapor deposition. *Adv. Mater.* **23**, 4898–4903 (2011).
87. Han, G. H. et al. Seeded growth of highly crystalline molybdenum disulphide monolayers at controlled locations. *Nat. Commun.* **6**, 6128 (2015).
88. Guo, Y. F. et al. Additive manufacturing of patterned 2D semiconductor through recyclable masked growth. *Proc. Natl Acad. Sci. USA* **116**, 3437–3442 (2019).
89. Guo, Y. et al. Designing artificial two-dimensional landscapes via atomic-layer substitution. *Proc. Natl Acad. Sci. USA* **118**, e2106124118 (2021).
90. Kim, K. S. et al. Non-epitaxial single-crystal 2D material growth by geometric confinement. *Nature* **614**, 88–94 (2023).
91. Tan, C. et al. 2D fin field-effect transistors integrated with epitaxial high-k gate oxide. *Nature* **616**, 66–72 (2023).
92. Zhao, M. V. et al. Large-scale chemical assembly of atomically thin transistors and circuits. *Nat. Nanotechnol.* **11**, 954–959 (2016).
93. Chiu, M. H. et al. Metal-guided selective growth of 2D materials: demonstration of a bottom-up CMOS inverter. *Adv. Mater.* **31**, 1900861 (2019).
94. Song, S. et al. Atomic transistors based on seamless lateral metal-semiconductor junctions with a sub-1-nm transfer length. *Nat. Commun.* **13**, 4916 (2022).
95. Wu, R. et al. Bilayer tungsten diselenide transistors with on-state currents exceeding 1.5 milliamperes per micrometre. *Nat. Electron.* **5**, 497–504 (2022).
96. Xu, M. et al. Reconfiguring nucleation for CVD growth of twisted bilayer MoS_2 with a wide range of twist angles. *Nat. Commun.* **15**, 562 (2024).

Acknowledgements

X.D. acknowledges support from the National Key R&D Program of the Ministry of Science and Technology of China (grant no. 2022YFA1203801), the National Natural Science Foundation of China (grant nos 51991340, 51991343 and 52221001) and the Innovative Research Groups of Hunan Province (grant no. 2020JJ1001). J.L. acknowledges support from the National Natural Science Foundation of China (grant nos. 52102168 and 52372145) and the Natural Science Foundation of Hunan Province (grant nos 2023JJ20009 and 2023RC3092). X.Y. acknowledges support from the National Natural Science Foundation of China (grant no. 62341402), Yongjiang Talent Introduction Programme (grant no. 2023A-167-G) and the Ningbo Natural Science Foundation (grant no. 2023J023).

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to Xidong Duan or Xiangfeng Duan.

Peer review information *Nature Materials* thanks Humberto Gutierrez and Vincent Tung for their contribution to the peer review of this work.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© Springer Nature Limited 2024

[1]Hunan Provincial Key Laboratory of Two-Dimensional Materials, State Key Laboratory for Chemo/Biosensing and Chemometrics, College of Chemistry and Chemical Engineering, Hunan University, Changsha, China. [2]Institute of Micro/Nano Materials and Devices, Ningbo University of Technology, Ningbo, China. [3]School of Physics and Electronics, Central South University, Changsha, China. [4]Department of Chemistry and Biochemistry, University of California, Los Angeles, Los Angeles, CA, USA. [5]These authors contributed equally: Jia Li, Xiangdong Yang. ✉e-mail: xidongduan@hnu.edu.cn; xduan@chem.ucla.edu